👩‍🏫 Staff Guide · Biology A · Unit 6 Blood Sugar & Diabetes (Level 2)

Level 2 = about 3rd-grade reading. SLE students stay in Ms. Dunham's lesson and use these pieces next to it: same topics, same order, same points.

1. How the pieces line up

Gen ed piece (Canvas: "Unit 3 Diabetes: Diabetes pretest, A1 What is going on (CDC data), A3/A6 Homeostasis (blood sugar balance game), ClEvR quizzes #1-3, A7 Genetic & Environmental Risks, A8 Solutions, A10 Diabetes Review, progress tracker, notebook check, Diabetes Test")	**Level 2 piece**
Diabetes pretest + A1 What is going on (CDC map)	Notes part A + slides (CDC map done with staff)
A3 / A6 Homeostasis + UW blood-sugar game + Homeostasis Gizmo	Notes part B + our blood-sugar order game (no login)
A7 Genetic & Environmental Risks (bean/coin simulation, online risk test, survey)	Notes part C; risk test done for a made-up person (see 💗)
A8 Solutions	Notes part D
Fed Up, That Sugar Film, Roots of Resilience	Film guides (L1 + L2) in this unit
Diabetes PSA poster + PSA source	PSA poster packet (🔬 Labs & Projects)
ClEvR quizzes #1-3, A10 review, Diabetes Test (28)	Modified test + key
Biggest Loser big paper / summary table / review	Not used (💗); review content folded into Units 4-6 quizzes
Class slides	Deck: 20 slides, Words to Know first, hold-up-1-2-3 checks, recap
Her worksheets (grouped into lessons)	Guided notes: 16 items + key
Class quiz / notes check	Quiz: 10 questions + key
Vocabulary	11 picture cards (cut apart)
—	2 games on the class site: matching + step-by-step order

2. Supports built in

- Short lines, key words bold, word bank in every notes box
- Quiz: 3 choices, no "all of the above"
- Picture anchors: 🔑 insulin = key, 🌡️ homeostasis = thermostat, 🩸 diabetes = sugar too high
- Level 1: no numbers from the CDC data; Level 2: 1-2 numbers only

3. Add these as the IEP allows

- Read-aloud · extended time · separate setting
- Printed slides · videos paused at the "look for" part

4. What to grade

- Test stands in for the paper Diabetes test (28 pts), scaled
- **Notes**: same points as her worksheets for that lesson when complete.
- **Quiz**: score, then scale to the class points.
- These are **modifications** (less content). The IEP must allow modified grading. Check with the case manager.

5. Who to ask

- Mr. Driver, Room 113 (SLE): changes and IEP questions.
- Facts we could not confirm in her files say "check with Ms. Dunham".

🚧 Under construction (check with Ms. Dunham)

- **Diabetes Test (28 pts, paper)**: Not in the export. Our test is built from her ClEvR quizzes and A3.10 review (labeled "From review").
- **Homeostasis Gizmo**: Needs her ExploreLearning login; no file. Our blood-sugar order game stands in.
- **Sweet berries / Miracle berry**: Title only in Canvas (a tasting?). Need directions and allergy rules.

💗 Sensitive topic note

Diabetes may be personal for students or families. Never ask students about their own or family health, weight, or blood sugar. The online risk test and survey in A7 ask for personal health information: do them for a made-up person ("Sam, age 45, sits most of the day, family history yes"). We dropped the "Biggest Loser" framing and all before/after body photos. Keep talk on bodies and communities, not blame.

6. Photo credits (open licenses)

Blood Glucose Testing.JPG — David-i98 (talk) (Uploads), CC BY-SA 3.0, Wikimedia Commons · Blausen 0699 PancreasAnatomy2.png — BruceBlaus. When using this image in external sources it can be cited as: Blaus, CC BY 3.0, Wikimedia Commons

👩🏾‍🏫 Staff Guide · Diabetes PSA Poster (Level 2)

Biology A · Unit 6 · goes with the class “Diabetes PSA poster + PSA source” (5 + 5 pts). Level 2 = about 3rd-grade reading. Same activity, same day, same points as the class.

1. The pieces

- **Poster card**: Diabetes PSA poster + PSA source: an 8.5×11 poster for Ferndale students about type 2 diabetes in another culture or country. 7 steps, one per box.
- **Prep sheet**: jobs, tools checklist, safety. Do it the day before or at the start.
- **Slides**: jobs, safety, steps, 2 checks.
- **Reflection**: replaces the written parts.
- **Game**: on the class site.

2. Prompts while they work

- “What place did you pick?”
- “Point to your title.”
- “What does insulin do?”

3. Modifications

- Source picked with staff
- Plan sheet with word bank

4. What to grade (5 + 5 pts)

- Same 5 + 5 points: her checklist
- Pointing, saying, or circling counts as a full answer.
- Modified grading only as the IEP allows. Check with the case manager.

5. Ask

- Mr. Driver, Room 113 (SLE).

💗 Sensitive topic note

Pick places and facts with respect. Students research a community, never their own or classmates' health.

Blood sugar notes (A1, A3/A6, A7, A8)

Biology A · Unit 6 Blood Sugar & Diabetes · Level 2 guided notes. Use with the class slides.

Name: _____

Fill in each blank with a word from the word bank in that box. Each word is used one time.

A. What is going on? (A1)

1. Most diabetes in adults is type _____.
2. Diabetes rates are higher in older adults and in people with less _____ or schooling.
3. Which is a fair CLAIM from the CDC data? (circle 1)

Many American adults are at risk for type 2 | Few Americans get type 2

Word bank: **2** | **money**

B. Homeostasis (A3 / A6)

4. Keeping the body in balance is called _____.
5. Insulin is made by the _____.
6. Insulin is like a _____ that lets glucose into cells.
7. Number what happens after a meal (1-4):

_____ pancreas releases insulin

_____ blood glucose goes back down

_____ blood glucose goes up

_____ glucose enters cells

Word bank: **homeostasis** | **key** | **pancreas**

C. Genetic and environmental risks (A7)

8. Risk that comes from your family is _____.
9. Match the risk factor to its type:

family history	environmental (lowers risk)
drinking soda every day	environmental
exercise	genetic

Word bank: **genetic**

D. Solutions (A8)

10. Drinking _____ instead of soda helps keep blood sugar steady.

11. Which are prevention ideas? (circle 2)

more PE time | healthy school lunches | more soda machines

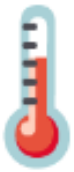
Word bank: **water**

Level 2 version of the gen ed "A1 What is going on, A3/A6 Homeostasis, A7 Genetic & Environmental Risks, A8 Solutions."

📖 Blood Sugar & Diabetes: Vocabulary Picture Cards (Level 2)

Name: _____

Cut on the dashed lines. Use the cards to match, sort, or quiz a partner.



Homeostasis

The body keeping inside conditions balanced, like a thermostat keeps a room just right.



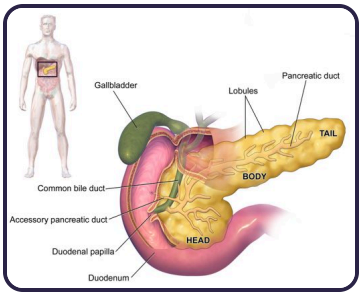
Blood sugar

Glucose in the blood. It goes up after eating.



Insulin

A hormone from the pancreas. Like a key, it unlocks cells so glucose can go in.



Pancreas

An organ near the stomach that makes insulin and glucagon.



Glucagon

A hormone from the pancreas that lets stored sugar out into the blood when blood sugar is low.



Diabetes

A disease where blood sugar stays too high for too long.



Type 1 diabetes

The pancreas makes little or no insulin. Not caused by food choices.



Type 2 diabetes

Cells stop answering insulin well, or not enough insulin. 90-95% of cases. Risk can be lowered.



Genetic risk

Risk from genes passed down from parents. We cannot change it.



Environmental risk

Risk from food, activity, sleep, money, and place. Some of it we can change.



Glucose meter

A tool that measures blood sugar from a tiny drop of blood.

📣 Diabetes PSA Poster

Biology A · Unit 6 · Level 2



Name: _____

Diabetes PSA poster + PSA source: an 8.5×11 poster for Ferndale students about type 2 diabetes in another culture or country.

🧺 Poster parts (her checklist)

- ☐ 🏷️ **1** catchy title
- ☐ ❓ **1** what type 2 diabetes is
- ☐ 🔢 **L1: 2 / L2: 5** facts about your culture or country
- ☐ 🖼️ **2** pictures or graphs

🌱 Steps: check each box when you finish

1. 🌎 **Choose** a culture or country (not Washington or high school students). ☐
2. 📰 **Find** one news source with staff. Paste the link on the source sheet. ☐
3. 🏷️ **Write** a catchy title. ☐
4. ❓ **Explain** type 2 diabetes in 1-2 sentences (insulin, blood sugar). ☐
5. 🔢 **List** 5 facts about diabetes in that place. ☐
6. 🖼️ **Add** 2 pictures or graphs. ☐
7. ✅ **Check** her list. Submit. ☐

📣 Poster plan

Culture/country: _____ Source: _____

Title: _____

Type 2 diabetes is _____ (word bank: insulin, blood sugar, cells, too high).

Facts: 1._____ 2._____ 3._____ 4._____ 5._____

📣 Diabetes PSA Poster · Prep Sheet

Biology A · Unit 6 · Level 2



Name: _____

👥 Job roles: circle your job. Initial when it is done.

Job	What you do	Done ✔️
🔎 **Fact finder**	Reads the source with staff and picks the facts.	
🖼️ **Picture picker**	Picks 2 pictures or graphs.	
✏️ **Writer**	Writes the title and facts.	
🎨 **Designer**	Places everything on the page.	

🧰 Tools and supplies: check when you have it

- ☐ 💻 **Slides or paper** 8.5×11 page
- ☐ 📰 **1 source** picked with staff from a news site (not YouTube, social media, or Wikipedia)
- ☐ 🖍️ **Markers** if on paper
- ☐ 📋 **This plan** fill it in first

⚠️ Safety and manners focus

📰 Use one good news source (USA Today, NY Times, Bellingham Herald…). No YouTube, social media, Wikipedia, Reddit.

🤝 Talk about a community with respect. No blaming.

🔒 No personal health facts about you or anyone you know.

📣 Diabetes PSA Poster · Reflection

Biology A · Unit 6 · Level 2



Name: _____

📸 My photos / proof

☐ Finished poster ☐ Source link

✅ Skills I used today (check all you did)

☐ 🏷️ Catchy title

☐ ❓ I explained type 2

☐ 🔢 5 facts from my source

☐ 🖼️ 2 images or graphs

⭐ What is one fact you want Ferndale students to know?

1. _____
2. _____

Ideas: **type 2** **risk** **healthy choices**

📰 Why is your source reliable?

1. _____
2. _____

Ideas: **news site** **checks facts**

✅ Blood Sugar & Diabetes Quiz

Biology A · Unit 6 · Level 2 (modified). 10 questions · 3 choices each · 1 point each.

Name: _____

Circle the best answer. You may have the questions read aloud.

1. 📊 Which is an accurate claim from the CDC data?

A. Few Americans develop type 2 diabetes

B. Many American adults are at risk for type 2 diabetes

C. Exercise increases the risk of type 2

2. 🏃 Exercise lowers the risk of type 2 diabetes because…

A. Jump rope is the only way

B. It makes you hungry

C. Exercise uses sugar, which lowers blood sugar

3. 🥗 Which claim fits the blood sugar data?

A. Healthy food choices can help regulate blood sugar

B. Food does not affect blood sugar

C. Bread does not raise blood sugar

4. 🔥 Low-intensity exercise lowers blood sugar because…

A. You stop eating

B. Muscles release sugar

C. Cells take in more sugar for energy

5. 🩸 Which is true of BOTH types of diabetes?

A. They cannot regulate blood sugar well with insulin

B. They make extra insulin

C. They are caused only by candy

6. 🏭 Insulin is made by the…

A. Liver

B. Pancreas

C. Stomach

7. 🔓 When blood sugar is low, the pancreas releases…

A. Insulin

B. Glucagon

C. Saliva

8. 🌡️ The body's ability to stay in balance is…

A. Homeostasis

B. Digestion

C. Combustion

9. 🧬 Which is a GENETIC risk factor?

A. Daily soda

B. Not sleeping

C. Family history

10. 🥗 Which is a prevention idea from A8?

A. More time for physical activity at school

B. Bigger soda sizes

C. Less PE

🎬 Fed Up (film)

Biology A · Unit 6 · Level 2 video guide (modified). Film: Fed Up (2014, PG). Staff shows the film; pause at the food-industry and sugar parts.

Name: _____

Before you watch

What do you think makes people's blood sugar go up?

_____

While you watch: circle the best answer

1. The film says the main problem in our food is too much…

A. Water | B. Added sugar | C. Vegetables

2. Insulin's job in the film: it helps the body…

A. Make bones | B. Digest protein | C. Store sugar as fat

3. Sugar in a piece of fruit is better than in soda because…

A. Fiber slows it down | B. Fruit has no sugar | C. Soda is cold

4. Food companies market a lot of their products to…

A. Only doctors | B. Farmers | C. Children

5. Why isn't exercise alone the answer, says the film?

A. You can't out-run a bad diet | B. Exercise is bad | C. Exercise raises blood sugar

After you watch: finish the sentence

6. One solution from the film:

One solution is _____ because _____.

_____

Goes with the class "A8 Fed Up questions" (5 pts). Same points when complete.

🎬 Roots of Resilience (Native communities and diabetes)

Biology A · Unit 6 · Level 2 video guide (modified). Video: Roots of Resilience: Confronting Diabetes in Native American Communities (YouTube, link in her Canvas).

Name: _____

Before you watch

What foods are traditional in your family or culture?

_____

While you watch: circle the best answer

1. What historical factors raised diabetes in Native communities?

A. Too many vegetables | B. Colonization and loss of traditional foods | C. Too much exercise

2. Which foods replaced traditional diets?

A. Fresh corn and beans | B. Wild rice and fish | C. Processed flour, sugar, lard, canned goods

3. Chef Sean Sherman ("The Sioux Chef") works to…

A. Bring back Indigenous foods | B. Sell soda | C. Close farms

4. Food sovereignty means a community…

A. Buys all food from far away | B. Stops eating | C. Controls its own healthy food

5. Why is diabetes both a medical and a cultural issue in the film?

A. It is tied to history, food, and family ways | B. It is only about medicine | C. It is only about sports

After you watch: finish the sentence

6. One personal action I could take:

I could _____. _____

Goes with the class "Roots of Resilience video worksheet (her answer key used)" (5 pts). Same points when complete.

🎬 That Sugar Film

Biology A · Unit 6 · Level 2 video guide (modified). Film: That Sugar Film (2014). Staff shows it (her Edpuzzle class link stays in her Canvas).

Name: _____

Before you watch

Which foods do you think have hidden sugar?

_____

While you watch: circle the best answer

1. Damon eats sugar that is ___ in foods that seem healthy.

A. Missing | B. Hidden | C. Frozen

2. Did Damon plan to keep the same amount of exercise?

A. No, he stopped exercising | B. He doubled it | C. Yes, he kept it the same

3. What happened to his liver?

A. It started to store fat | B. It got smaller | C. Nothing

4. What did the milkshake do to his brain?

A. It made him sleepy only | B. Nothing | C. It lit up the reward part

5. How did he feel after he stopped eating so much sugar?

A. Better | B. The same | C. Worse

After you watch: finish the sentence

6. The independent variable in Damon's experiment was…

The independent variable was _____. _____

Goes with the class "A9 That Sugar Film questions (Edpuzzle in her class)" (5 pts). Same points when complete.

🚧 Homeostasis Gizmo

Biology A · Unit 6 · Level 2 · Under construction

Name: _____

🚧 **Under construction:** The Gizmo needs her ExploreLearning class login and there is no worksheet in the export. *Check with Ms. Dunham.*

What we have so far

Use our Blood Sugar step-by-step game (insulin and glucagon) instead.

Staff: until this is ready

Have the student do the matching Level 2 unit game or vocabulary cards for this lesson, and give the same points for effort. Mr. Driver, Room 113.

🚧 Sweet Berries / Miracle Berry

Biology A · Unit 6 · Level 2 · Under construction

Name: _____

🚧 **Under construction:** Canvas has only the titles (the Miracle berry page is empty). It may be a tasting lab. We need the directions, the food, and allergy rules. *Check with Ms. Dunham.*

Staff: until this is ready

Have the student do the matching Level 2 unit game or vocabulary cards for this lesson, and give the same points for effort. Mr. Driver, Room 113.

Blood sugar notes (A1, A3/A6, A7, A8) · ANSWER KEY

Biology A · Unit 6 Blood Sugar & Diabetes · Level 2 guided notes. Use with the class slides.

Name: _____

Fill in each blank with a word from the word bank in that box. Each word is used one time.

A. What is going on? (A1)

1. Most diabetes in adults is type **2**.
2. Diabetes rates are higher in older adults and in people with less **money** or schooling.
3. Which is a fair CLAIM from the CDC data? (circle 1)

Many American adults are at risk for type 2 | Few Americans get type 2

Word bank: 2 | money

B. Homeostasis (A3 / A6)

4. Keeping the body in balance is called **homeostasis**.
5. Insulin is made by the **pancreas**.
6. Insulin is like a **key** that lets glucose into cells.
7. Number what happens after a meal (1-4):

2 pancreas releases insulin
4 blood glucose goes back down
1 blood glucose goes up
3 glucose enters cells

Word bank: homeostasis | key | pancreas

C. Genetic and environmental risks (A7)

8. Risk that comes from your family is **genetic**.
9. Match the risk factor to its type:

(Key: family history → genetic · drinking soda every day → environmental · exercise → environmental (lowers risk))

family history	environmental (lowers risk)
drinking soda every day	environmental
exercise	genetic

Word bank: genetic

D. Solutions (A8)

10. Drinking **water** instead of soda helps keep blood sugar steady.

11. Which are prevention ideas? (circle 2)

more PE time **healthy school lunches** more soda machines

Word bank: **water**

Level 2 version of the gen ed "A1 What is going on, A3/A6 Homeostasis, A7 Genetic & Environmental Risks, A8 Solutions."

🔑 *Key written by our team from Ms. Dunham's materials, please check.*

✅ Blood Sugar & Diabetes Quiz · ANSWER KEY

Biology A · Unit 6 · Level 2 (modified). 10 questions · 3 choices each · 1 point each.

Name: _____

Circle the best answer. You may have the questions read aloud.

1. 📊 Which is an accurate claim from the CDC data? CIEvR #1 Q1

- A. Few Americans develop type 2 diabetes
- **B. Many American adults are at risk for type 2 diabetes**
- C. Exercise increases the risk of type 2

2. 🏃 Exercise lowers the risk of type 2 diabetes because… CIEvR #1 Q3

- A. Jump rope is the only way
- B. It makes you hungry
- **C. Exercise uses sugar, which lowers blood sugar**

3. 🥗 Which claim fits the blood sugar data? CIEvR #2 Q1

- **A. Healthy food choices can help regulate blood sugar**
- B. Food does not affect blood sugar
- C. Bread does not raise blood sugar

4. 🔥 Low-intensity exercise lowers blood sugar because… CIEvR #2 Q3

- A. You stop eating
- B. Muscles release sugar
- **C. Cells take in more sugar for energy**

5. 🩸 Which is true of BOTH types of diabetes? CIEvR #3 Q1

- **A. They cannot regulate blood sugar well with insulin**
- B. They make extra insulin
- C. They are caused only by candy

6. 🏭 Insulin is made by the… A3 Homeostasis

- A. Liver
- **B. Pancreas**
- C. Stomach

7. 🔓 When blood sugar is low, the pancreas releases… A3 Homeostasis

- A. Insulin
- **B. Glucagon**
- C. Saliva

8. 🌡️ The body's ability to stay in balance is… A3.10 review

- **A. Homeostasis**
- B. Digestion
- C. Combustion

9. 🧬 Which is a GENETIC risk factor? A7 Risks

- A. Daily soda
- B. Not sleeping
- **C. Family history**

10. 🥗 Which is a prevention idea from A8? A8 Solutions

- **A. More time for physical activity at school**
- B. Bigger soda sizes
- C. Less PE

Answers

1-B · 2-C · 3-A · 4-C · 5-A · 6-B · 7-B · 8-A · 9-C · 10-A

Reworded from the gen ed "ClEvR Quizzes #1-3 (Canvas, answers stored), A3 Homeostasis, A3.10 review, A7, A8". Score: __ / 10.

🔑 *“ClEvR” items use the answers stored in her Canvas quizzes (reworded, choices cut to 3). Other items: key written by our team, please check. Her Diabetes test (28 pts) is on paper and not in the export; this is a “From review” test.*

🎬 Fed Up (film) · ANSWER KEY

Biology A · Unit 6 · Level 2 video guide (modified). Film: Fed Up (2014, PG). Staff shows the film; pause at the food-industry and sugar parts.

Name: _____

Before you watch

What do you think makes people's blood sugar go up?

_____

While you watch: circle the best answer

1. The film says the main problem in our food is too much…

A. Water | **B. Added sugar** | C. Vegetables

2. Insulin's job in the film: it helps the body…

A. Make bones | B. Digest protein | **C. Store sugar as fat**

3. Sugar in a piece of fruit is better than in soda because…

A. Fiber slows it down | B. Fruit has no sugar | C. Soda is cold

4. Food companies market a lot of their products to…

A. Only doctors | B. Farmers | **C. Children**

5. Why isn't exercise alone the answer, says the film?

A. You can't out-run a bad diet | B. Exercise is bad | C. Exercise raises blood sugar

After you watch: finish the sentence

6. One solution from the film:

One solution is _____ because _____.

_____

Goes with the class "A8 Fed Up questions" (5 pts). Same points when complete. Her sheet has 38 questions, many about people's weights. We kept 5 content questions + 1 solution question and left out all weight questions.
💗 The film shows kids' weights and weight struggles. We left out every weight question. Don't ask students about their own weight.
🔑 Key written by our team from Ms. Dunham's materials, please check.

🎬 Roots of Resilience (Native communities and diabetes) · ANSWER KEY

Biology A · Unit 6 · Level 2 video guide (modified). Video: Roots of Resilience: Confronting Diabetes in Native American Communities (YouTube, link in her Canvas).

Name: _____

Before you watch

What foods are traditional in your family or culture?

_____

While you watch: circle the best answer

1. What historical factors raised diabetes in Native communities?

A. Too many vegetables | **B. Colonization and loss of traditional foods** | C. Too much exercise

2. Which foods replaced traditional diets?

A. Fresh corn and beans | B. Wild rice and fish | **C. Processed flour, sugar, lard, canned goods**

3. Chef Sean Sherman ("The Sioux Chef") works to…

A. Bring back Indigenous foods | B. Sell soda | C. Close farms

4. Food sovereignty means a community…

A. Buys all food from far away | B. Stops eating | **C. Controls its own healthy food**

5. Why is diabetes both a medical and a cultural issue in the film?

A. It is tied to history, food, and family ways | B. It is only about medicine | C. It is only about sports

After you watch: finish the sentence

6. One personal action I could take:

I could _____. _____

Goes with the class "Roots of Resilience video worksheet (her answer key used)" (5 pts). Same points when complete. Answers match the answer key in her Roots of Resilience worksheet file.

💗 This film is about Native communities and historical harm. Speak about communities with respect; do not single out any student's background.

🔑 Key written by our team from Ms. Dunham's materials, please check.

🎬 That Sugar Film · ANSWER KEY

Biology A · Unit 6 · Level 2 video guide (modified). Film: That Sugar Film (2014). Staff shows it (her Edpuzzle class link stays in her Canvas).

Name: _____

Before you watch

Which foods do you think have hidden sugar?

_____

While you watch: circle the best answer

1. Damon eats sugar that is ___ in foods that seem healthy.

A. Missing | **B. Hidden** | C. Frozen

2. Did Damon plan to keep the same amount of exercise?

A. No, he stopped exercising | B. He doubled it | **C. Yes, he kept it the same**

3. What happened to his liver?

A. It started to store fat | B. It got smaller | C. Nothing

4. What did the milkshake do to his brain?

A. It made him sleepy only | B. Nothing | **C. It lit up the reward part**

5. How did he feel after he stopped eating so much sugar?

A. Better | B. The same | C. Worse

After you watch: finish the sentence

6. The independent variable in Damon's experiment was…

The independent variable was _____. **the amount of sugar he ate**

Goes with the class "A9 That Sugar Film questions (Edpuzzle in her class)" (5 pts). Same points when complete. Her sheet has 24 questions with many exact numbers (teaspoons, kilos, prices). We kept the main ideas and the variables question; number answers should be checked with her key.

💗 The film tracks one man's weight gain. We left out the weight numbers. Don't ask students about their own weight.

🔑 Key written by our team from Ms. Dunham's materials, please check.